

INHA UNIVERSITY IN TASHKENT
School of Computers and Information Engineering
Spring semester 2025
Assignment #2 [Computer Networks]

Name:- _____ ID: _____ Group: _____

Submission Date: 07 /03/ 2025 [Friday] (23:30 Hr)

NOTE :

1. Computer typed solution are not acceptable (**only handwritten** allowed)
2. Take the **print of this sheet** and solve on it.
3. Submissions not conforming to the given format will not be evaluated.

Q1. Complete the following table for IPv4 classful addressing

Class	Leading bits	% allocation of address space	No. of networks	No. of Hosts / Network	Default mask
Class A					
Class B					
Class C					
Class D					

Q2. An organization is granted block 130.56.0.0/16. The administrator wants to create 1024 subnets.

- a. Find the subnet mask.
- b. Find the number of addresses in each subnet.
- c. Find the first and the last address in the first subnet.

Answer

Calculations

- a. subnet mask _____
- b. no. of addresses _____
- c1. 1st address in 1st subnet _____
- c2. last addr. in 1st subnet _____

Q3. Find the range of addresses in the following blocks:

a. 123.56.77.32/28

b. 180.34.64.64/29

Answer

a. 123.56.77.32/28

b. 180.34.64.64/29

Q4. In classless addressing, if we know the first and the last address in the block. Can we find the prefix length? If the answer is yes, show the process and give an example.

Answer